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Chapter 10

Workshop Management Overview



Introduction

Workshop Management is a comprehensive control tool which greatly simplifies the administration of activities concerned with the repair and overhaul of equipment at the service provider's premises.

Workshop Management is a complementary product to Service Management, both of which can run concurrently using the same data structure, and both are fully integrated with System 21 Financial and Inventory modules.

It supports multi-branch operations within each of a number of companies. It also supports contract maintenance, from quotations, pricing and renewal to producing maintenance invoices, including in advance, arrears or by instalment.

Job processing is controlled by call logging, work control and planned maintenance. You can also escalate jobs. Full enquiry features mean you can access master files, transaction history, job status, and financial data.

Where System Manager is installed, you can restrict access to certain functions by operator ID, or create tailor-made menus for each operation. This means you can prevent operators from accessing specific functions, or information belonging to certain companies.

Relationship to Other Modules

Workshop Management operates under the control of System Manager. The full benefits of the system are gained by integrating it with the following prerequisite modules:

- Accounts Receivable
- General Ledger
- Inventory Management
- Sales Order Processing.

You can also integrate it with the following:

- Advanced Financial Integrator (AFI)
- Distribution Requirements Planning
- World Trade
- Forecasting.

Workshop Management Configuration

Company Level Information

You can operate Workshop Management for a number of companies, maintaining the characteristics of each on a control file. You control Workshop by setting up data for the company and then for each branch within the company. You normally do this after creating the Accounts Receivable, Inventory Management and Sales Order Processing companies.

You define certain data and policies at the company level. These include:

- Document reference numbers: invoice, credit note, contract, job
- General Ledger accounts
- Default labour price list
- Default scheduled visit profile
- Working day start / finish times
- Hours entry only, for Job Reporting.

You also define service and accounting periods at company level. You can enter up to 99 service periods per year. They are mainly used to schedule planned maintenance visits. You can define each day of the year as a working or non-working day, so you can calculate target response date / time for jobs.

You can enter up to 99 accounting periods per year. These should be consistent with those specified in the ledgers. The number of days in the accounting periods is used to apportion deferred revenue from contracts invoiced in advance.

Branch Level Information

Once you have defined the control parameters for a company, you must define at least one branch. A branch is a physical or logical location that has responsibility for servicing specific customer sites.

You can add workshop branches after starting Workshop Management.

You define Certain data and policies at the branch level. These include:

- Branch name and address
- Printer output queue for job audit log
- Job escalation parameters: escalation time fence
- Escalation reporting interval
- Users to alert at each escalation level
- Default response hours.

You must authorise each user to access at least one branch within a company, but you can authorise the user to several or all branches. This authorisation by branch restricts the data to which a user has access.

Reference Data

Workshop Management uses a number of reference files that hold data which rarely changes:

- Standard codes and descriptions
- System parameters
- Customers
- Divisions, model groups, model sub-groups and models
- Installations
- Contract types
- Job categories
- Labour rate price lists
- Service territories, teams and field service groups
- Engineers
- Contract rates
- Scheduled visit profiles.

Standard Codes and Descriptions

You use the descriptions, or codes / parameters file to set up codes with standard descriptions. At data entry this validates the information and displays descriptions on both screens and reports. In some specific instances, you can store a parameter or number, for example value added tax percentages.

For Workshop Management to work correctly, certain standard codes and parameters must be present and these are supplied with the system.

Setting Up Customers and Equipment

Before you can record contract information or enter service operations, you must enter customer details, location and the equipment installed at the sites.

You must set up:

- Customer accounts, maintained through Accounts Receivable.
- Installation addresses, maintained through Accounts Receivable.
- Installed equipment details.
- Equipment models.
- Model group and sub-group maintenance.



Note: Some of this information is also used by Accounts Receivable and Sales Order Processing. Make sure that data is correct for all purposes.

You can print and enquire on all the information.

Before you can set up customer and installation details in Workshop Management you must enter the customer details using the Names and Addresses option in Accounts Receivable and the Installation Address Maintenance, in Accounts Receivable and Sales Order Processing.

Customer Name and Address Details

You enter additional customer details for Workshop Management using the Customer Additional Details Maintenance option.

You cannot enter installation records or contracts, or process transaction, without customer accounts. Although Workshop Management does not use some of the customer account data, the following parameters are significant:

- Name and address - Displayed on many screens and printed on some reports, including invoices / credit notes.
- Geocode or postcode - Used to automatically assign teams of engineers to jobs, by means of the geocode / territory definitions and engineer team / field service group / territory assignment rules.
- VAT code and indicator - Used within invoice / credit note generation to establish which VAT rate to charge.
- Credit stop indicator - Used within job maintenance to provide warning that the customer is on credit stop.

Installation Address Details

An installation address is a customer address at which equipment is installed. A customer account address can be an installation address, and there can be several additional installation addresses for the customer account.

When you set up a customer account, an address code of 000 is added. You can then create further installation addresses by assigning address code suffixes of 001, 002, and so on to each address record for the customer.

Each installation address is identified by a unique combination of account code and address code.

You link additional customer details required by Workshop Management to account and installation addresses.

You must create the site address before entering the details of equipment installed at the address. You cannot perform any contract or transaction processing until this equipment database is present. The following data for an installation address is significant:

- Name and address - Displayed on many screens and printed on some reports, including invoices / credit notes.
- Postcode - Used to automatically assign engineers to jobs at the customer site, by means of the postcode, or geocode / territory definitions and engineer / territory assignment rules.

Price List Options

The two price list options are both Sales Order Processing options, see SOP product information for more details.

Customers

Workshop Management shares a common customer database with Accounts Receivable and Sales Order Processing. Accounts Receivable controls credit limits and payment terms, Sales Order Processing controls stock allocation, pricing of sold items and Sales Analysis coding, and Workshop Management controls pricing of service items and invoice consolidation levels.

You can manipulate the customer information to match the structure of the customer's organisation. You can have a simple single account, or a statement account with many Invoice To accounts, each with many different delivery or installation accounts.

Statement Account

This is shared with Accounts Receivable and is normally created and maintained by Accounts Receivable. The data includes credit details, Accounts Receivable parameters and statistical codes.

Invoice Account

You can send invoices to an invoice only address.

Delivery or Installation Account

Each account can have many delivery or installation addresses. Define these in Sales Order Processing or Workshop Management or both, by adding a three-character address code suffix to the account code.

Additional Information

You must set up additional information, for each account address, where you service equipment. Workshop Management uses this information and it includes the invoice destination address, invoice consolidation level (account, address code, or contract / job), service region, and price list for service parts.

You can identify a customer either by entering the account and address, or by using the available name and address search routines.

Models

You identify each piece of equipment defined to Workshop Management by its model code and, normally, its serial number. You can use standard model codes and numbers to describe the different types or configurations of equipment. Link each model to a model sub-group and link the sub-group to a model group.

You can specify models as individual machines, or as peripherals to a machine.

You can hold contract rates for different types of service contract at model group, model sub-group or model code level. You can assign engineers to only work on specific model groups, model sub-groups or models.

You can determine the labour rates to charge by the model on which the work is performed, since each model code points to the appropriate labour price list.

Installations

You define an installation within Workshop Management as a customer address at which equipment or systems are installed.

You identify the customer address by an account code and an address code. The site can have many items of equipment or systems installed. You should set up installation records for all installations irrespective of whether they are covered by service contracts.

You identify each piece of equipment at an installation by the unique combination of model number and serial number. If you do not know serial numbers, then use the model number and quantity.

The data held for each piece of equipment includes details of age, delivery and installation dates, warranty period, location, department, complexity points and product division.

Engineers

You must define each engineer who services or repairs equipment on the engineer master file. Details include name and address, team code, grade, hourly cost rate, service region, stockroom code, and skills matrix.

The stockroom code is the link to Inventory and must be defined in Inventory Management. It represents a discrete area within a company where you record and control stock separately from other company stocks. This stockroom could be a van or the workshop stores.

Territories And Engineer Assignment

Workshop Management can automatically assign engineer teams to service jobs, if you have:

- Defined the customer's geographic or postcode on the customer record.
- Divided the geographic area covered by the company into territories; defined by a list of full or partial geocodes or postcodes.
- Specified the engineers' teams who cover each territory.
- Defined the field service group for the team, to show which product divisions, model groups, model sub groups and models the team services. The field service group record contains a skills matrix to define these limits.

Workshop Management can map the customer's address to a service territory and allocate work to the team of engineers designated to operate in that territory.

You can automatically or manually allocate an individual engineer within the team to a specific job. In job selection, you can further specialise according to the engineer's skill level, that is you can assign an engineer to service specific model groups (and optionally sub-groups) or to perform work of a specific type (job category), or to perform work on equipment belonging to a specific division or any combination of any of the above.

Contract Types

You can use contract types to define the different types of service contract offered by the company to its customers. Each contract type record holds the normal duration of the contract, guaranteed response hours, invoice schedule (in advance, in arrears, or on completion of each scheduled maintenance visit), and whether you generate a quotation or contract at renewal.

You can apply billing parameters to the contract type. These indicate if service charges, rental charges, or various meterage conditions apply.

All contracts have a user-defined contract type code. Workshop Management defaults a contract type of *NO for work on equipment not covered by a service contract.

The parameters set up for the contract type are defaults for the contract, but you can change them, if required.

Job Categories

You use job categories to describe the different types of work required. You can define a number of categories as defaults for breakdowns, warranty and repeat calls to save time when you log a call.

You must also define a table of valid job categories for each contract type. This table specifies whether you charge the customer for: travel and labour hours (fixed or hourly charges), up to four parts groups and miscellaneous charges, when you do work of a certain job category on equipment covered by that contract type.

Price Lists

Workshop Management use two types of price list; for labour rates and for parts.

Labour rate price lists define the hourly labour rates charged for contract, non-contract and workshop work. The price lists include an hourly travel rate and effectivity dates, and are linked to models. The model code of the equipment determines the labour rates charged.

Parts price lists and discount lists define the prices and discounts charged for spare parts fitted or used by an engineer. You maintain these lists, which also have effectivity dates, in Sales Order Processing. You link them to the customer additional details records in Workshop Management: the prices charged are determined by the customer site at which the work is done.

Contract Rates

You define contract rates for each contract type at model group, model sub-group, or individual model code level.

Contract rates specify the charge per scheduled maintenance visit and a fixed charge for each piece of equipment covered by the contract type. Contract rates have effectivity dates to control new rates.

Contract pricing routines use contract rates, but only when the system has been defined for term processing; not for monthly processing.

Scheduled Visit Profiles

You use scheduled visit profiles to define standard schedules for planned maintenance and service visits to equipment under contract. You define the schedules in terms of the number, type and frequency of visits.

You can associate a visit profile with a contract, model, or model group. You can use them to automatically generate the service visit schedule for any equipment added to a service contract.

Contract Processing

Workshop Management provides flexible and comprehensive facilities for contract management, as follows:

- Maintenance
- Pricing and invoicing
- Renewal
- Credits.

Contract Maintenance

You identify service contracts by a contract number, type and start date. Each contract relates to one or more locations of a single customer, and can cover one or more pieces of equipment installed at these locations.

When you enter a new contract, you create contract header and equipment records.

The contract header record includes the contract duration and end date, order reference and date, and invoice destination. The service parameters include response hours, scheduled visit profile, if there is cover while the contract is pending, and the rates applied. The billing parameters set the invoice term, frequency and whether charges are in advance, in arrears or per visit. The service conditions provide for setting up service, visit and rental charges at contract header level: the service and visit charges override the contract rates set by model group, model sub-group or model; the service conditions can be overridden at equipment and peripheral levels.

You set up the contract equipment records once the header is validated and saved. You select items, from a list of all the equipment defined for the customer account code, to place. You can only select equipment previously specified as resident at a location to which the contract applies. You can add service conditions at equipment and peripheral levels. You can create for each item on a contract, a schedule of planned service visits, by service period for the duration of the contract. You can create this automatically using the service visit profile associated with the contract header, model, or model group, and then amend the schedule as required.

Contract Pricing And Invoicing

Workshop Management prices and invoices contracts invoiced in advance or in arrears as follows:

Service Pricing

- A contract is priced for a defined period of time (invoice term) which can be the same or shorter than the contract duration. The charge can be split into instalments for invoicing; for example you can invoice a one year term in quarterly instalments. Workshop Management pro-ratas the charge for equipment added to a contract part-way through its invoice term.
- To price a contract, each piece of equipment is priced individually and accumulated to give a total contract price. If equipment and its peripherals have special prices, then these are the prices charged.
- If no special price exists, then Workshop Management uses the contract rates file (term processing only), to get the price per maintenance visit and any supplement fee. The total number of scheduled maintenance visits in the invoice term is multiplied by the price per visit and the supplement fee added to get the gross charge for the piece of equipment.



Note: Service contracts can include fixed visit charges, invoiced after each visit. You invoice these visits through job invoicing, not contract invoicing.

Rental Pricing

- The rental charge can be on the contract header, this is the default price for each item added to the contract, or can be maintained on the contract equipment line. The charge entered will be per month if you selected monthly processing on the system parameter file, or per term if you selected term processing.

Meterage Pricing

- You can only use meterage charges with monthly processing.
- You can define a minimum volume on the contract header or equipment line. The minimum monthly charge is the minimum volume multiplied by the price per copy, which you can invoice in single or multiple months.
- You can define a minimum charge on the contract header or equipment line, this may be invoiced in single or multiple months.
- You can define up to five copy charge bands, for each of two meters, for each contract equipment line.
- Use interim pricing to charge for a meter estimate, or actual reading returned by the user, based on the minimum units already charged for, (or actual units used). Define the frequency of interim pricing on the contract header.

- Use reconciliation pricing to charge according to an actual meter reading. You can apply additional meterage charges or give credits for overcharges. The reconciliation term must be a multiple of the interim period.
- You can pool meterage pricing, that is you can spread the minimum volumes or values across all equipment on the contract, so the excess units on one machine can be offset against the under-usage on another machine.

Invoicing

- Contract pricing creates records on an invoices pending file. The generate and print invoices option generates invoice header and line records. Invoice consolidation, which groups invoice pending lines into invoices, takes place as specified on the customer additional details records, that is by account, by account and location or by contract. VAT is calculated using the parameters set on the customer account record. The invoices are then printed.
- Invoice generation posts the transactions to the sales and general ledgers. If you invoice in advance, the revenue value of the invoice is deferred in the General Ledger, by generating additional postings to transfer period revenue values from a deferred income account into a sales account each period.

Contract Invoice Term

When pricing a contract, the price is calculated to cover a fixed period of time known as the Contract invoice term. There can be one or many invoice terms within the duration of a contract.

The Contract invoice term is the period of time from the Next Term Starts On Date to the Next Term Ends On Date, and defines the period of price protection for the contract.

Next Term Starts On Date

The date on which the next Contract invoice term starts. Contracts are selected for pricing if this date is less than or equal to the date entered on the Price contracts prompt screen. Maintain this field on the billing parameters window.

Next Invoice Term Ends On Date

The date on which the next contract invoice term ends. Maintain this field on the billing parameters window.

Normal Invoice Term

The normal invoice term for the contract. Maintain this field on the billing parameters window.

Invoice Frequency

A sub-division of the normal invoice term. This is the period of time, in months, between invoices, if the contract is invoiced in instalments. Maintain this field on the billing parameters window.

Contract Payable in Advance

A contract for which contract charges are invoiced before the term or instalment.

Contract Payable in Arrears

A contract for which contract charges are invoiced after the term or instalment.

Contract Payable Per Visit

A contract for which contract charges are invoiced as a result of a scheduled service visit. These visits are priced by the Price Jobs option.

Pending Invoice Line

An invoice line, not yet assigned an invoice number or posted to the Ledgers, waiting to be grouped with other invoice lines to form an invoice. The grouping depends on the invoice consolidation level, set up for the customer account (000 record) using the Additional Service Details option.

Pending Invoice Line Availability Date

The date on, or after which, the pending invoice line may be included in an invoice. For advance contracts, this is the start date of the contract invoice term or instalment period; for arrears contracts, this is the end date of the contract invoice term or instalment period.

Description

You can select all branches or a single branch, specify the type of billing, and enter a cut-off date relating to the contract invoice terms. This only processes contracts payable in advance or arrears.

Pending invoice lines are generated for:

- All non-invoiced pieces of equipment, added to a contract during an invoice term, which have been priced.
- All equipment on contracts where the entered cut-off date is later than, or equal to, the Next Term Starts On Date of the contract.

Pricing Routine

Contract pricing is governed by the conditions set up in contract maintenance, at header or equipment levels, or at both.

Default Prices from the Contract Rates File



Note: You can only use default pricing for fixed service, rental and visit charges. It does not apply to copy / vend contracts, where you must set up special contract conditions.

If you do not overwrite the Service Conditions at contract header or contract equipment levels for the Fixed Service, Rental or Visit Charges, they default from the Contract Rates file. Contract pricing selects the fixed service, rental and visit charges from the Contract Rates file, for the appropriate effective date, equipment group, contract type, term and currency.

For a particular currency, the contract rates record used is for the contract type and term, effective on the Pricing date. The Pricing date is derived as follows: if the Use start date rates field on the contract header is set to 1=Yes, or if there is a Price held until date which is later than the Next term starts on date, the pricing date is the contract start date; otherwise, the pricing date is the Next term starts on date.

The equipment model group, model sub-group and model are compared to the same fields on the contract rates record; the record with the best match will be selected. If no record is found, the contract could be left with no charge: special

service conditions could be set up by the user for the contract. The recommendation is that records should be created on the Contract Rates file for as many combinations of equipment and contract types as are necessary.

The price is calculated, by multiplying the visit charge by the number of scheduled maintenance visits in the Contract invoice term, for this piece of equipment.

The fixed service charge, from the contract rates record, is added; the charge applies to the Normal invoice term, and is adjusted to allow for the Contract invoice term: the number of days in the Contract invoice term is divided by the number of days in the Normal invoice term, and this factor is applied to the fixed service charge.

The rental for the contract term is included, but separately from the service and visit charges.

Contract Conditions Prices at Header / Equipment Levels

If you enter a price on the Contract Conditions Window at header or equipment levels, this price applies to the Normal invoice term specified on the contract. The Normal invoice term is converted to days, the Term Ends On date and the number of days between these two dates is calculated. The number of days in the term is also calculated. The ratio of the latter number of days to the former is calculated and applied to the contract conditions price, to get the actual price for the contract, or for the specific piece of equipment, for the contract invoice term.



Note: For complete instalment periods, the invoices are of equal value.

Pro-Rata Pricing

The price for equipment added or removed from a contract is calculated on a pro-rata basis.

The number of days in the Normal invoice term of the contract is calculated, as described above.

For equipment added late, the contract invoice term is the number of days between the date the equipment was added and the term ends date, or equipment removal date, whichever comes first.

For equipment removed early, the contract invoice term is the number of days between the date the equipment was added or the term start date, whichever comes last, and the term end date, or equipment removal date, whichever comes first.

The ratio of the two sets of days (that is contract invoice term and normal invoice term) is calculated and applied to the calculated price for the normal invoice term, to get the actual price for the contract, or for the specific piece of equipment, for the number of days that it has been on the contract.



Note: Equipment removed, or scheduled for removal, within a processed invoice term will not have an automatic credit generated; for this, use the Contract Credit option.

Contract Renewal

You can run this function at any time. It prompts for a date representing the expiry date limit of contracts for renewal. The system reads through all active contracts and determines which ones to renew, by comparing the contract end date with the entered date. If the contract end date is less than or equal to the entered date, the contract is renewed.

In order to renew a contract, the equipment on the contract must still be eligible for the contract type, that is the contract charge rates must be in effect.

A new set of contract records will be generated with either the same contract number or a new, system-assigned contract number. The contract start date will be the day after the existing contract end date. Other contract details are copied to the new contract.

The status of new contracts will be set as either pending start date or quotation, depending on the Quotation Required At Contract Renewal flag on the contract type record. You can amend the contract details generated (through contract maintenance procedures) before generating quotations or invoices.

Contract Credits

You can process credits at any time, to refund part, or all, of a contract invoice, and to cancel any pending invoice records. You can select the record to amend by contract, invoice, or period.

You can amend the contract and invoice record to affect the operative date and values involved. You can discard the selected record, or give a full or partial credit, or credit the original charge and re-invoice a different amount.

To produce the resultant invoices and credit notes, select the Generate and Print (Contract) Invoices option. This procedure also removes any cancelled pending invoice records from the work file.

Job Processing

An important function of any service organisation is to complete, in an efficient and timely manner, both customer service requests and planned maintenance work according to the contractual terms agreed.

When you have completed a service jobs, you want to invoice any chargeable work promptly to optimise cash flow.

Workshop Management provides the following facilities to help meet these objectives:

- Planned Maintenance Job Generation.
- Call Logging.
- Work Control.
- Engineer Diary.
- Technical Reporting.
- Job Pricing and Invoicing.
- Job Escalation.

Planned Maintenance Job Generation

Run this process as required. Load PM Jobs creates a service job for each installation that has a planned maintenance visit scheduled for periods within a specified range. You can create schedules for up to six periods in advance of the current period. The service schedule details, held for equipment covered by contract, provide the base data for processing.

The process attempts to assign an engineer team to each job, by comparing the customer geographic code with the territory definitions, and the model type requiring service with the field service group that covers the division, model group, model sub-group and model. If you entered a specific engineer against the equipment on the installation record, this engineer will always be assigned (if this method of selection is defined on the system parameter file).

If more than one piece of equipment at an installation address, on the same contract, requires a service visit in the same period, then this procedure normally generates only one job covering all the equipment. However, you can set an option on the company profile to generate an individual job for each piece of equipment.

Call Logging

Use this facility to record unscheduled or breakdown calls and generate service jobs. You can also amend the details of outstanding jobs.

Addition of New Job

To log a new job you must enter customer account details, the service contract number, the serial number, or the model number and serial number. You can only log a new job if an installation record exists for the customer address requesting service. You can create installation details for non-account customers.

If the customer is on credit hold or stop, a warning message displays and the call status set to Credit Hold, but you can still enter the job.

You must select each item to include on the job from the displayed list of equipment. If the customer has a number of items of equipment on site and cannot identify which of them requires the service, You can log the job against any available machine line. When the engineer completes the job you must enter the true model and serial number.

Workshop Management checks for previous jobs (for qualifying job categories), to determine if this is a repeat call. This compares the number of days or meter units or both elapsed since the previous call, with the values defined on the volume segment file for the model. Workshop Management allocates a job number. The response hours are calculated by reference to the special serial numbers file, the contract line, the contract header, or the contract type (possibly reduced by the 3-D Matrix percentages). A target start date and time is set by adding the retrieved hours to the date and time the call is logged.

The job category defaults to the value entered in job category maintenance. Enter the customer contact name and fault code and description. You can change the response hours, and therefore the target response times. Alternatively, you can enter an appointment date and time. You can also assign an engineer to the job.

Amendment to an Existing Job

You can select the job you want to amend by customer, serial and model number, and contract type or number.

You can amend the job category, engineer assigned, customer contact and fault code for each piece of equipment on the job.

Work Control

You can either use the Work Control option to manually assign engineers, or get Workshop Management to automatically assign engineers using the automatic call allocation program.

If you require automatic call allocation you must start the job and make sure the Workshop Management sub-system is running.



Note: The sub-system library list must be tailored to take account of: the use of consolidated files libraries in place of IN, OE, SL and G files libraries and the version of Business Manager in use, for adjusting the four IPG libraries.



Tip: If you are running System 21 Version 3 on an AS/400, then use the CHGJOB command and **F4**, enter OSLSSP/OSLS2P3, then select **F10** and page down to the library list.

Automatic call allocation is a sleeper job, that automatically schedules engineers within a team, to the jobs assigned to the team in job logging. Assignment is on the basis of available engineer, based on current workload and skill profile, and who can achieve an estimated time of arrival equal to or less than the target date and time for the call. If the selection criteria cannot be met, an error message is displayed for the call, and a non-assignment reason is entered on the job detail screen.

You can schedule jobs manually, in which case the same criteria and checks for engineer selection are applied.

Use the Work Control function to despatch outstanding jobs to engineers and enter status updates on engineers' current jobs.

You can display a list of jobs assigned to a specific engineer or team of engineers, or for the branch. You can extract planned or unscheduled jobs, and jobs of certain status. The list of jobs display in order of target date and time, with the most urgent jobs first.

You can display jobs for a team can graphically: colour blocks represent each job and the colour denotes the urgency of the job. Unscheduled jobs for the team display in the lower part of the screen; scheduled jobs display under each engineer in the upper part of the screen. You can move jobs, and display job line details for team and engineer work queues.

Engineer Technical Reporting

Use this facility to enter the details of completed work. You access it by selecting the completed job line with 99.

A technical report is made up of four sections:

- Header Details - engineer number, job number, visit date, report reference, time on site, customer and engineer travel time.
- Labour Hours - hours spent on each piece of equipment, qualified by fault code, job category, machine section and sub-section repaired, and corrective action. You can book hours to a basic rate and three overtime rates.
- Parts Used - Spares parts and quantities used, by job category.
- Miscellaneous Costs - analysed by job category, you enter a cost or charge type, the cost value and charge value or enter a percentage uplift to derive the charge.



Note: Both parts and labour can be chargeable (100%) or non chargeable (0%) or any percentage in between.

If a job requires more than one engineer to complete the work, you can enter several technical reports. The prime report has the suffix-00, with each subsequent report suffix incremented by the system. After you enter the final report, indicate the job is finished and fully documented. At this point, you can price and invoice the job.

Job Pricing And Invoicing

Workshop jobs, not fully covered under the terms of a service contract, must be charged to the customer. You price and invoice the jobs after you complete them through the technical reporting function.

The decision as to which jobs, or elements of jobs, are chargeable is made by referencing the user-defined matrix of contract types and job categories. For the particular combination, the matrix indicates whether labour or travel distance or travel time or miscellaneous costs or four types of parts or any combination of the above, are chargeable to the customer.

Six types of charge can appear on a job invoice:

- **Contract Visit Fee** - Contracts can include fixed visit charges to invoice after each visit: the contract's billing parameters being set accordingly. You invoice these visits through job invoicing at job completion. The fixed visit charge is obtained from the contract conditions at equipment or header levels, or from the contract rates file.
- **Labour** - The labour charge is calculated from the number of hours booked on the engineer report, multiplied by the hourly rate on the labour price list for the model. The hourly rate used will be either a contract, or a non-contract, rate depending on whether the piece of equipment was covered by contract when the job was logged. The hourly rate will be uplifted by a factor for overtime rates. You can apply a fixed labour charge, if required by the job's charge matrix set up in Cover Type / Job Category Maintenance.
- **Travel Labour** - The travel time is calculated from the number of customer travel hours and minutes booked on the engineer's technical report, multiplied by the hourly travel rate from the labour price list held for the model. A fixed travel labour charge may be applied, if required by the job's charge matrix set up in Contract Type / Job Category Maintenance.
- **Parts** - The charge for parts used by the engineer is calculated according to the price list and discount list defined for the customer site. The price and discount list details are obtained from Sales Order Processing.
- **Miscellaneous Costs** - The charges are based on the values entered through the technical reporting function.
- **Chargeable Travel** - The charges may be based on a standard distance value, the actual distance driven, or zones. Data is set up on the Additional Service Details record for each site. All jobs, even non-chargeable ones, should be processed through the job pricing and invoicing functions, since the cost values of labour, engineer travel time, miscellaneous costs and parts, for all jobs, are calculated by these routines.

A batch job is submitted by this menu option to price jobs for the current branch. Only job lines which have been fully technically reported are processed by this function.

Pending invoice lines are generated for all job lines which have been fully technically reported but not yet invoiced. If a job bill to account exists, this is used to generate the pending invoice. If there is no job bill to account but there is an

ownership account, this is used instead. If there is no ownership account, the site account is used.

The price charged for each job line is built up from the following five individual charges:

Labour Charge

Each labour hours entry from each technical report for the job line is priced individually.

A labour charge is only made, if the contract type and job category combination is defined as chargeable for labour, on the cover type / job category file.

If you have defined fixed charges, they are applied before any variable labour charges. Variable charges are applied on the time remaining, after any fixed charges. The labour hours, on each technical report entry, are multiplied by the premium factor (from the codes file LHTY record) for the labour hours type. The result is multiplied by an hourly labour rate from a labour rate price list for the appropriate currency, which is derived as follows:

The labour rate price list is taken from the customer additional details, then the model file. The price list contains four, hourly labour rates: contract, non-contract, workshop and travel. If the engineer is a workshop engineer, that rate is used; otherwise, the contract or non-contract rate is used; and if travel time is involved, the travel rate is used.

In identifying the actual price list record to be used, the pricing date is taken into account, that is the price list record effective at the pricing date will be used. The pricing date is determined as follows:

- If the equipment is not covered by contract: Pricing date=Visit date (from technical report)
- If the Use start date rates field on the contract which covers the equipment is set to 1=Yes: Pricing date=Contract start date
- If the Use start date rates field on the contract, which covers the equipment, is set to 0 (No), and the job is a planned (scheduled) job, and the visit date is before the Price held until date on the contract which covers the equipment, then: Pricing date=Contract start date
- If not, or if Price held until date is zero: Pricing date=Visit date
- If the job is a callout (breakdown) job and the call logged date is before the price held until date on the contract which covers the equipment, then: Pricing date=Contract start date
- If not, or if Price held until date is zero: Pricing date=Call logged date

The labour cost value is always calculated, whether or not the labour is chargeable. The cost value, in base currency, is the number of labour hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer who performed the work). No premium factor is applied to the cost rate.

Parts Charge

A charge for a part is only made if the contract type and job category combination is defined as chargeable for parts on the cover type / job category

file. When reporting parts used on a job, the charge % may be reset from 0% to any value up to 100%.

Parts are priced by retrieving the parts price, discount list, and currency codes held on the Additional Service Details record. Using the price and discount list records, defined in the Sales Order Processing application (and effective for the pricing date, as derived for the labour pricing calculations), the prices and discounts are calculated in the appropriate prime currency.

The cost value of parts used is always calculated, whether or not the parts are chargeable. Parts costs are always set up in Inventory Item / Stockroom Details in base currency. The cost value is calculated in base currency, not within the Price Jobs function, but when the Inventory Management application is updated with the detail of the parts used.

Miscellaneous Charge

A miscellaneous charge is only made if the contract type and job category combination is defined as chargeable for miscellaneous charges on the cover type / job category file.

The charge value of each miscellaneous charge line is the charge value entered on the line itself in the specified currency.

The cost value of each miscellaneous charge is always calculated, whether or not the item is chargeable. The cost value is the cost value entered on the line itself in the specified currency. If the specified currency is not the base currency, the system automatically converts the prime currency value to base.

Travel Charge

Fixed charges: these are applied prior to any variable travel charges, if defined on the cover type / job category file.

Variable charges are applied on the time remaining after any fixed charges.

The prime currency travel charge is calculated as the billable customer travel time entered on the technical report, multiplied by the travel rate obtained from the labour rates price list. No premium factor is applied to this rate.

The base currency travel cost value is always calculated for engineer travel time. The cost value is the number of travel hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer on the technical report).

Contract Visit Charge

If the job line being invoiced is part of a planned maintenance job and the contract under which the job is performed is currently defined as being invoiced per visit a contract visit charge is generated (unless calculated as zero).

The contract visit charge is calculated as described in the previous section Price Contracts: the number of visits covered by the job is multiplied by the visit fee. The visit fee may default from the contract rates file, or be specified as a contract condition at header and / or equipment line levels.

Engineer Diary

Workshop Management provides an engineer diary, which enables the planning and scheduling of non-machine-related jobs and other commitments for each engineer. Job appointments are automatically entered in the diary for the engineer when you log a call or in work allocation.

The next five working days are displayed, starting from the system date. Each day in the diary is free format; bookings can be entered for any date, time and duration, or for a date and time from with a date and time to.

Diary entries are made as required and are used to reserve the times for events such as holidays, training courses, sickness, dental appointments, etc. As entries for engineers are scheduled for a date and time, they will appear both in the engineer's work queue and in the graphical display of his jobs.

The diary is accessible from the Work Control option.

Job Escalation

Workshop Management can alert you to any jobs approaching their designated response time and still outstanding. Job escalation runs so long as the Workshop Management sub-system is active.



Note: The sub-system library list must be tailored to take account of the use of consolidated files' libraries in place of IN, OE, SL and GL files' libraries and the version of System Manager used for adjusting the four IPG libraries.

All service jobs created are stamped with a target response date and time, which is calculated as follows:

Call-out Jobs

For jobs entered using the Job Logging / Maintenance function the target response date and time is:

Date / time call logged + response hours

Where response hours is derived from the special serial numbers file, or the contract if a contract exists. If the equipment is not covered by a contract, the response hours default to the value for the contract type *NO. You can make manual overrides of the response hours if special circumstances arise.

When calculating target response date and time, Workshop Management takes into account working and non-working days, and the standard service window for the company, that is hours service is provided. These are all user-defined.

Planned Jobs

For jobs generated from the service schedules defined for contracts, the target response date and time is calculated as:

Target date = Date of last working day of the period in which the job is scheduled

Target time = Start time of the working day

You can define the escalation time fence, the interval between each escalation step, and the users alerted at each step. Workshop Management monitors each outstanding job, comparing its target date and time with the current system date and time. Once the job enters the defined escalation time fence, it starts to escalate and with the escalation step recalculated at frequent intervals: the escalation step moving from high to low value as the job approaches its target date and time. Escalation step zero shows that the job has reached, or gone beyond, its target time.



Note: Workshop Management assumes a 7-day working week. If you work less than this, Workshop Management escalates Monday's work on a Sunday, not on the preceding Friday.

As a job moves from one escalation step to the next, Workshop Management can send a message to pre-defined users to alert them so they can take appropriate action.

Workshop Management removes jobs from the escalation procedure once they reach a certain status. You define these within the Job Line Status Code Record. You can view the escalation status of jobs using various selection parameters.

Example of Call Escalation

Escalation Step	Reporting Times - (1 Hour Intervals)	
0	16:00	Target Time
1	15:00	
2	14:00	
3	13:00	
4	12:00	Escalation Time Fence (4 hours) starts

The escalation time fence is set to 4 hours and the reporting interval to 1 hour.

A job with target time of 16:00 starts to escalate at 12:00.

Between 12:00 and 12:59, the escalation step is 4, because the job is within 4 hours of target. Between 13:00 and 13:59, the escalation step is 3, because the job is within 3 hours of target, and so on.

At 16:00, or later, the escalation step is 0, because the job has reached or passed its target time.

Warranty Claims

This is a financial module attached to Workshop Management that you can use to create credit notes for dealers because of warranty claims against the supplier. This module uses all the customer, parts and models information you entered in Workshop Management.

If a customer can make warranty claims, you must set the Warranty Claim flag on the customer additional details to either parts, labour or both parts and labour. You also set the labour rate list to use for the warranty claim in the customer additional details.

You can use this function to:

- Enter the warranty claim data, such as model, fault, purchase date, customer details, dealer information.
- Record any parts used during the repair.
- Validate the data entered.
- Accept the claim and automatically create a Sales Order Processing credit note for labour or parts or both.
- Reject the claim with the appropriate reason and produce a letter stating the reason for the rejection.
- Maintain the credit status of the customer making the claim.

The validation process checks all the claim details and recommends if the claim should be accepted, rejected or whether you need further information before you can process the claim further.

This function also provides facilities to analyse the warranty claim data. You can:

- Search for other faults and repairs on this model and serial number.
- Check if parts have been claimed for before using the same invoice number.
- Display and audit all stages that the claim has passed through.

Once you have accepted or rejected a claim, Workshop Management archives the claim.

Enquiries

Workshop Management provides a comprehensive range of enquiries for:

- Master Files - These generally show the same details as their counterpart maintenance routines.
- Transaction History - The job history enquiry shows details of historical service work performed against equipment. The invoice enquiry shows details of a workshop invoice, whether it be a contract, job, or sundry invoice.
- Status - An escalation enquiry shows up-to-date information on the escalation status of workshop jobs.

Loading Workshop Management Data

You must load the following data for each workshop company you want to process within the live module.

The data is divided into two groups:

- Basic data - data used mainly for reference purposes, which is low in volume and not subject to ongoing change, such as codes.
- Major data - data which constitutes the bulk of the module and is subject to regular change, such as customers and contracts.

Some of the data listed is marked Optional. This means that, in system terms, the data is not essential to running the module; but in practice, the data may well be essential, for example, contracts.

Basic (Parameter) Data

- Branch Records - one for each service branch
- Calendar Control Record
- Codes / Parameter File Records - System-required parameters have a * at the beginning of the Parameter Type description: these must be set up, or copied from the demonstration data supplied. In addition, set up appropriate code values for each of the following types:
 - ABSC (Engineer absent)
 - CCRC (Contract cancellation reasons)
 - CHGT (Miscellaneous costs / charges)
 - CORA (Corrective actions)
 - DAYT (Day types)
 - EGRD (Engineer job grades)
 - FLTC (Faults)
 - LHTY (Labour hours or overtime types)
 - RRES (Credit release reason)
 - STAT (Equipment status)
 - TXTO (Text destinations)
- Company Profile
- Contract Type Records - one for each contract type code
- Contract Rates Records (optional)
- Cover Type / Job Category Records - one for each combination of cover type, job category, (and Currency where World Trade and Multi-Currency are active) with invoicing indicators
- Daily Calendar File (& Rebuild Period End Dates) - for all years over which current service contracts are spread
- District - one for each work controller, to authorise work on teams
- Division Records - one for each service division
- Engineer Records - one for each workshop engineer
- Escalation Parameters (optional)
- Field Service Group - one for each group or equipment (division / model family / model to be serviced)
- Job Category Records - one for each job category code
- Labour Rates Price List Records - one for each set of labour rates (and for different currencies where World Trade and Multi-Currency are active)

- Model Group / Sub-Group Records - one for each model group / sub-group combination
- Model Records - one for each model code
- Scheduled Visit Profile Records (optional) - one for each standard schedule of planned maintenance visits
- System Parameters File
- Team - one for each engineer team
- Tax Codes (for World Trade and Multi-Currency only.) - for each country's contract type / job category combinations
- User / Branch Authority Records - one for each user and branch combination
- Volume Segment - minimum one, optionally more for each model, or group of models
- Zone Charge (optional) - one for each zone (and currency where World Trade and Multi-Currency are active)
- Inventory descriptions file
 - DEPT (Departments)
 - MANT (Mantle groupings)
 - MFID (Manufacturer's ID)
 - * - Mfr not specified
 - OTHR (Customer search parameters)
 - PTYP (Item type)
 - 4 codes are essential to categorise parts:
 - C - Consumables
 - J - Major spares
 - N - Minor spares
 - S - Supply items
 - 2 codes assist analysis of machine sales:
 - M - Machine sale via SOP
 - P - Peripheral sale via SOP
 - SPED (Urgency indicator)
 - STKU (Stockroom usage code)
 - B - Service spares stockroom
 - W - Service workshop: company stock
 - X - Service workshop: customer stock

Major Data

- Contract Records (optional) - one for each active service contract, with details of each piece of equipment covered and service schedules.
- Customer Account Records - one for each customer account.
- Customer Additional Details (Workshop Management) Records - one for each customer site.
- Installation Address Records - one for each site address, within a customer account.
- Installation Records - for each customer site, details of each piece of equipment at the site.